

J. ROY

LEAD FULL STACK PYTHON DEVELOPER

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SUMMARY

Full-Stack Python Developer with 11 years of experience in designing, developing, and optimizing high performance applications across web, desktop, data science, and cloud environments. Proficient in a diverse range of Python frameworks, libraries, and tools, with deep expertise in modern frontend technologies and scalable backend architectures. Skilled in leading end-to-end software development lifecycles, modernizing legacy systems, and building maintainable, scalable applications. Recognized for delivering robust data driven solutions, seamless system integrations, and workflow optimizations by leveraging Python's versatility and power. Passionate about collaboration, automation, and performance optimization to drive innovation and efficiency in software development.

TECHNICAL SKILLS

Technical Expertise

- Programming & Web Development: Python (2.x–3.x), JavaScript, TypeScript, HTML5, CSS3
- Frameworks & Libraries: Django, Flask, FastAPI, Pyramid, Tornado, Bottle, Sanic, CherryPy, React, Angular, Vue.js, Flask Bootstrap, Jinja2 Templates
- API Development: RESTful APIs, GraphQL, gRPC, WebSockets, Socket.IO, Celery, Falcon, Connexion

Data, AI & Cloud Technologies

- Data Science & Machine Learning: NumPy, Pandas, Dask, SciPy, TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, Hugging Face, NLTK, spaCy, Gensim
- Big Data & Cloud: Apache Spark (PySpark), Hadoop, Apache Kafka, Apache Airflow, Snowflake, Vaex
- Databases: PostgreSQL, MySQL, SQLite, MongoDB, Cassandra, Redis, DynamoDB, SQLAlchemy

Automation, DevOps & Tools

- Testing & QA: PyTest, Unittest, Nose, Behave, Lettuce, Pylint, Flake8, Black, MyPy
- Automation & Web Scraping: Selenium, Scrapy, BeautifulSoup, Puppeteer, PyAutoGUI
- DevOps & CI/CD: Git, GitHub, GitLab, Bitbucket, Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps
- Development & Collaboration Tools: PyCharm, VS Code, Jupyter Notebook, Spyder

PROFESSIONAL EXPERIENCE

Full-Stack Python Developer | Tetra Tech | Remote

Jul 2022 – Present

- Designed and deployed high-performance RESTful APIs using Django REST Framework and Flask, processing 10M+ monthly transactions.
- Developed scalable microservices with FastAPI and Docker, improving response times by 40%.
- Built real-time data dashboards with Plotly Dash and Flask for business intelligence.
- Integrated AI-driven recommendation engines into cloud services using AWS Lambda and S3.
- Led monolithic-to-microservices migration, ensuring seamless Kubernetes deployments.
- Automated CI/CD pipelines with GitHub Actions and Jenkins, streamlining testing and deployment.
- Mentored junior developers in Python best practices, test-driven development, and data processing.
- Managed cloud infrastructure with AWS (EC2, S3, Lambda, RDS) for high availability.
- Implemented OAuth2, JWT authentication, and role-based access control (RBAC) for API security.
- Optimized database performance with PostgreSQL, MySQL, and MongoDB, ensuring efficient queries.
- Designed asynchronous task processing with Celery & Redis, improving background job execution.

Senior Python Developer | Tetra Tech | Remote

May 2019 – Jun 2022

- Led Django + React e-commerce platform, boosting engagement by 25% and efficiency.
- Optimized backend with Celery & RabbitMQ, improving reliability by 40%.
- Built RESTful & GraphQL APIs with Django & FastAPI, cutting response times by 35%.
- Developed ETL pipelines with Apache Airflow, reducing latency by 50%.
- Automated CI/CD with GitHub Actions & Jenkins, cutting release cycles by 60%.
- Migrated monolithic apps to microservices with Docker & Kubernetes for scalability.
- Optimized PostgreSQL & MySQL queries, enhancing data retrieval efficiency.
- Secured APIs with OAuth2, JWT, and RBAC, ensuring industry compliance.
- Mentored junior developers, leading code reviews and training sessions.

Python Software Engineer | Genesis IT Lab

Jun 2016 – Apr 2019

- Developed a logistics tracking system with Flask & MongoDB, improving efficiency by 30%.
- Built SOAP & RESTful APIs, enhancing system interoperability by 40%.
- Optimized data processing with NumPy, Pandas & Airflow, reducing latency by 50%.
- Improved PostgreSQL & MySQL queries, boosting performance by 35%.
- Developed NLP models using NLTK & spaCy, increasing classification accuracy by 25%.
- Automated background tasks with Celery & Redis for efficient job execution.
- Saved 500+ hours annually by automating recurring tasks with Python scripts.
- Wrote unit & integration tests with PyTest, achieving 90% code coverage.

Junior Python Developer | Airbnb

Jun 2013 – May 2016

- Contributed to a Django-based client management system, improving onboarding workflows.
- Developed and maintained RESTful APIs using Django and Flask for seamless data exchange.
- Enhanced frontend responsiveness using AJAX, jQuery, and Flask backends.
- Automated data extraction and web scraping using BeautifulSoup and Selenium.
- Created batch-processing scripts for financial data, optimizing reporting efficiency.
- Designed and optimized SQL queries for PostgreSQL and MySQL, improving database performance.
- Built data exploration prototypes using Jupyter Notebooks for data analysis and visualization.
- Assisted in API security implementation using OAuth2 and JWT authentication.
- Developed reusable Python scripts for automating recurring tasks and data processing.
- Wrote unit and integration tests using PyTest, improving code quality and reliability.
- Assisted in code reviews, enforcing PEP 8 compliance and improving documentation.
- Collaborated with senior developers to debug and optimize application performance.

EDUCATION

Bachelors of Science Computer Science

Oct 2009 - Oct 2013

- University of Engineering and Technology

ADDITIONAL INFORMATION

- Strong foundation in Python Full-Stack Development, Data Science, and AI-driven solutions.
- Extensive experience in API design, microservices, DevOps, and performance optimization.
- Proven ability to lead projects, mentor teams, and deliver scalable solutions.
- Experience working in high-profile organizations like Airbnb, Tetra Tech, and Genesis IT Lab.
- Strong emphasis on automation, CI/CD pipelines, and cloud-based deployments